Metastable State

It is the state in which the liquid below its freezing point is in the supercooled state which is not very much stable.

$$\text{Supercooled liquid} \rightleftharpoons \text{Vapour}$$

On adding nucleus of ice (stable state), the system reverts to true stable system

$$\text{Solid} \rightleftharpoons \text{Vapour}$$

The Sulphur System

It is a one component, four-phase system. The four different phases are:

(i) Rhombic sulphur (S_R)
(ii) Monoclinic Sulphur (S_M)
(iii) Liquid Sulphur (S_L)
(iv) Vapour Sulphur (S_V)

As all the four phases can be represented by only one chemical entity 'Sulphur' (S), it is a one component system.

From the phase rule when C = 1,

$$\begin{aligned} F &= C - P + 2 \\ &= 1 - P + 2 \\ &= 3 - P \end{aligned}$$

As degree of freedom *can not have a minus value*, therefore out of four possible phases, only *three* are present at a time. It is not possible for a single component system to have four phases together at equilibrium. The degree of freedom for different cases will be:

when $P = 1, F = 3 - P = 3 - 1 = 2$ (bivariant system)
$P = 2, F = 3 - P = 3 - 2 = 1$ (monovariant system)
$P = 3, F = 3 - P = 3 - 3 = 0$ (nonvariant system)

The two crystalline forms of sulphur are rhombic sulphur and monoclinic sulphur. S_R and S_M exhibit enantiotropy* with a transition point at 95.6°C. Below 95.6C S_R is stable and above this temperature S_M is the stable variety. At 95.6°C each form can be gradually transformed to another form and the two forms remain in equilibrium. S_M melts at 120°C

Thus, $$S_R \xrightleftharpoons{95.6°C} S_M \xrightleftharpoons{120°C} S_L$$

The number of phases at any instant depends upon the temperataure-pressure conditions at that instant. The P – T diagram of sulphur system showing various equilibriums is given Fig. 2.3.

The phase diagram consists of :

(a) Four areas
  (i) Area ABG (Sulphur rhombic)
  (ii) Area BEC (Sulphur monoclinic)
  (iii) Area GECD (Sulphur liquid)
  (iv) Area ABCD (Sulphur vapour)
(b) Six curves AB, BC, CD, BE, CE, EG
(c) Three triple points B, C and E.

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*Enantiotropy: In some cases one polymorphic form (or allotrope) can change into another at a definite temperature when the two forms have a common vapour pressure. The temperature is known as the transition temperature. One form is stable above this temperature and the other form below it. When the change of form to the other at the transition temperature is reversible, the phenomenon is called enantiotropy and the polymorphic forms enantitropes. For example,

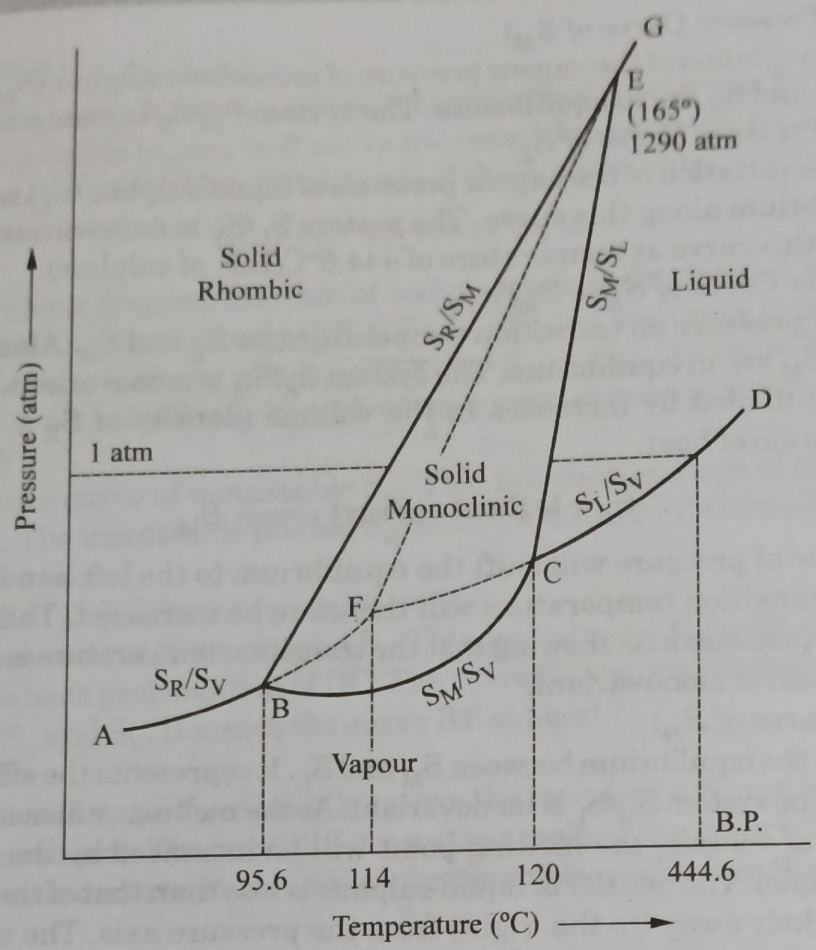


Fig. 2.3 Sulphur system

Areas

The phase diagram of sulphur system consists of four areas or regions. Each area represents the single phase system namely, rhombic sulphur, monoclinic sulphur, liquid sulphur and vapour.

For each area C = 1 and P = 1, and degree of freedom will be:

$$F = C - P + 2$$
$$= 1 - P + 2$$
$$= 2 \quad \text{(Bivariant system).}$$

Thus it is clear that each of the systems S_R, S_M, S_L and S_V are bivariant systems. It means that in order to locate any point in any of these area, the variables pressure and temperature both are required to be specified.

The Curves

The six curves AB, BC, CD, BE, CE, EG divide the diagram into four areas.

Curve AB (Vapour Pressure Curve of S_R)

Curve AB is the vapour pressure curve of S_R at different temperatures. Along this curve the two phases rhomic sulphur (S_R) and sulphur vapour (S_V) are in equilibrium. The system S_R/S_V has one degree of freedom.

$$F = C - P + 2$$
$$= 1 - 2 + 2$$
$$= 1 \quad \text{Mono variant system.}$$

Curve BC (Vapour Pressure Curve of S_M)

Curve BC shows the variation of the vapour pressure of monoclinic sulphur (S_M) with temperature. Along this curve S_M and S_V are in equilibrium. The system S_M/S_V is monovariant.

Curve CD (Vapour Pressure Curve of S_L)

Curve CD depicts the variation of the vapour pressure of liquid sulphur (S_L) with temperature. S_L and S_V are in equilibrium along this curve. The system S_L/S_V is monovariant. One atmospheric pressure line meets this curve at temperature of 444.6°C(B.P. of sulphur).

Curve BE (Transition Curve of S_R to S_M)

It shows the effect of pressure on transition temperature for S_R and S_M. Along this curve the two solid phases S_R and S_M are in equilibrium. The system S_R/S_M is monovariant. The transformation of S_R to S_M is accompanied by increases in the volume (density of S_R = 2.04 and density of S_M = 1.9) and absorption of heat.

$$S_R + Q \text{ (heat energy)} \rightleftharpoons S_M$$

Thus the increase of pressure will shift the equilibrium to the left hand side (Le Chatelier's Principle) and the transition temperature will therefore be increased. That is why the line BE slopes away from the pressure axis showing that the transition temperature is raised with increases of pressure. The system is monovariant.

Curve CE (Fusion Curve of S_M)

Curve CE represents the equilibrium between S_M and S_L. It represents the effect of pressure on the melting point of S_M. The system S_M/S_L is monovariant. As the melting or fusion of S_M is accompanied by a slight increase of volume, the melting point will be increased by the increase of pressure (Le Chatelier's Principle). The density of liquid sulphur is less than that of the monoclinic solid and hence CE slopes slightly away (to the right) from the pressure axis. The curve ends at point E because monoclinic sulphur (S_M) ceases to exist beyond this point.

Curve EG (Fusion Curve of S_R)

The two phases S_R and S_L coexist in equilibrium along this curve. S_R/S_L is monovariant.

Triple Points

In the sulphur system there are three triple points B, C and E. All the triple points represent the existence of three phases in equlibrium with each other, therefore;

$$\begin{aligned} F &= C + P + 2 \\ &= 1 - 3 + 2 \\ &= 0 \end{aligned}$$

The system corresponding to each point is nonvariant.

(i) *Triple Point B:* At this point the three curves AB, BC and BE meet. Three phases, S_R, S_M and S_V are in equilibrium at this point B. As, P = 3 and C = 1, the system is nonvariant. At this point B (95.6°C) S_R changes to S_M and the process is reversible.

(ii) *Triple Point C:* The three curves BC, CD and CE meet at this point. The three phases S_M, S_L and S_V are in equilibrium at this point. As P = 3 and C = 1, the system is nonvariant. The temperature corresponding to C is 120° C (melting point of S_M)

(iii) *Triple Point E:* The three curves CE, BE and EG join at this point. The two lines CE and BE have different inclinations away from the pressure axis. The system $S_R/S_M/S_L$ is nonvariant at the point E. This point E shows the conditions of existence of the $S_R/S_M/S_L$ at 155°C and 1290 atm pressure.

Metastable Equilibria

The transformation of S_R to S_M is a slow process. If enough time is not given for the change and S_R is heated rapidly, it is possible to pass well above the transition point (95.6°C) without obtaining monoclinic sulphur. In that case, there is existence of three phase only.

$$S_R \rightleftharpoons S_L \rightleftharpoons S_V$$

In this condition, phase diagram like that of water system is obtained which consists of three areas (AFG, GFD and DFA), three curves (BF, CF and FE) and one triple point.

In each of three areas, a single phase exists (S_R, S_L or S_V). As there being P = 1, C = 1 therefore, $F = C - P + 2 = 1 - 1 + 2 = 2$. In each of the areas, system is bivariant.

The Dashed Curve BF

It is the vapour pressure curve of metastable S_R. This is a continuation of the vapour pressure curve AB to stable S_R. The metastable phases S_R and S_V are in equlibrium along this curve. The system S_R/S_V is monovariant.

The Dashed Curve CF

This is the vapour pressure curve of supercooled S_L. This dashed curve CF is obtained on supercooling liquid sulphur. It is the back prolongation of DC. The curve CF depicts the metastable equilibrium between supercooled S_L and S_V. It meets the curve BF at point F.

The Dashed Curve FE

It is the fusion curve of metastable S_R. Along this curve the two phases S_R and S_L are in equilibrium and the system is monovariant. The curve FE shows that the melting point of S_R is increased with pressure. Beyond E, the curve shows the conditions for the stable equilibrium S_R/S_L as the metastable S_R disappears.

The Metastable Triple Point F

The three metastable phases S_R, S_L and S_V are in equilibrium at this point. The system $S_R/S_L/S_V$ is a non variant one at triple point. The temperature corresponding to triple point is 114°C (melting point of metastable S_R).

2.5 TWO COMPONENT SYSTEM

According to phase rule, the degree of freedom of two component system is given by:

$$\begin{aligned} F &= C - P + 2 = 2 - P + 2 \\ &= 4 - P \end{aligned}$$

When a single phase is present in two component system, the degree of freedom is three.

$$\begin{aligned} F &= C - P + 2 \\ &= 2 - 1 + 2 \\ &= 3 \end{aligned}$$

It means that three variables are required to explain the system completely. Thus in such a system, in addition to pressure and temperature the third variable, i.e. concentration of one of the substance has also to be specified. For representing these three variables graphically, three co-ordinate axis at right angles to each other would be required. Therefore a solid model of phase diagram will be formed. Such a phase diagram cannot be conveniently shown on paper.

But in actual practice, a simple plane diagram with two variables shown at two axis at right angle to each other is considered. It is customary to hold one of them constant. In this way three types of diagrams can be constructed, which are as follows.

(i) Pressure-Temperature diagram when concentration variable is kept constant. (P-T diagram)
(ii) Pressure-Concentration diagram when temperature is kept constant. (P-C diagram)
(iii) Temperature -Concentration diagram when pressure is kept constant. (T-C diagram)

Generally in most of the cases pressure is kept constant at 1 atmosphere and thus it is convenient to use temperature – concentration diagram.

The P – C diagram with constant temperature is known as *isothermal diagram* while the P – T diagram keeping composition constant is known as *isoplethal diagram*.

From the above discussion it is clear that for the sake of having simple plane diagrams we generally consider only two variables, third one being a constant. For example, for a solid-liquid equilibrium, usually the gas phase is absent and effect of pressure on the system is very small which is negligible, hence it is important to take into account the remaining variables like temperature and concentration and the effect of pressure can be discarded. **Such a solid liquid system with gas phase absent is known as a condensed system.**

The experimental measurements of temperature and concentration are usually carried out under atmospheric pressure. Since the degree of freedom in such a system is reduced by one (by keeping pressure constant), the reduced phase rule equation is as follows:

$$F = C - P + 2 \quad \text{(Phase rule equation)}$$

$$F = (C - P + 2) - 1 \quad \text{(Phase rule equation, modified for condensed system)}$$

$$F' = C - P + 1 \quad \text{(Reduced phase rule)}$$

where F' shows the remaining degree of freedom of the system. Reduced phase rule is widely applicable to solid-liquid two component reduced systems.

In a two component system the various possible phase equilibria are:

Liquid $\rightleftharpoons$ vapour

Solid $\rightleftharpoons$ liquid

Liquid $\rightleftharpoons$ liquid

Solid $\rightleftharpoons$ vapour

Solid $\rightleftharpoons$ solid

Here we shall confine our discussion to

Solid $\rightleftharpoons$ Liquid equilibria

2.6 SOLID–LIQUID EQUILIBRIA

As is discussed earlier that measurement on solid-liquid in condensed systems is carried out at atmospheric pressure. Since the degree of freedom in such a case is reduced by one (as pressure is kept constant) the modified phase rule for such systems is

$$F = C - P + 1 \qquad \text{....(ii)}$$

This is know as condensed phase rule. Now for *two component systems* equation (ii) becomes

$$F = 2 - P + 1 = 3 - P$$

When $P = 1$, then $F = 2$

Thus it has two variables (temperature and composition) which have to be specified. Therefore, solid liquid equilibria are best represented by temperature-composition diagram.

CONSTRUCTION OF PHASE DIAGRAMS

To construct the phase diagrams of a two component, solid-liquid equilibria it is necessary to determine the composition of the mixture at different temperatures. Various experimental techniques are there for this purpose but the most generally used one is the *thermal analysis*.

Thermal Analysis

A convenient method to study solid/liquid equilibria is through thermal analysis. In this method, solids of different compositions are separately heated above their melting points. The resulting liquids are cooled slowly and cooling curves are constructed by plotting temperature against time.

From the analysis of cooling curve for any mixture of a definite composition, it is possible to determine its

(i) Freezing point
(ii) Eutectic point

(i) **Analysis of the cooling curves**

(a) **Pure substance**. When a pure substance is cooled slowly and the temperature is noted at definite intervals, a curve of the type shown in Fig. 2.4 is obtained. Point 'b' denotes the temperature at which the liquid starts solidifying or freezing. After this the temperature remains constant till whole of the liquid is completely solidified (point c). Thereafter, the fall in temperature will again become continuous as is denoted by the curve cd. This curve represents the cooling of the solid mixture.

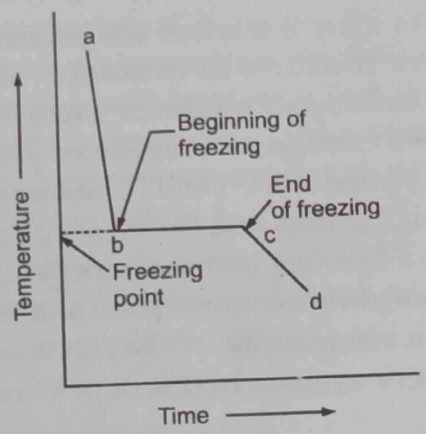


Fig. 2.4 Cooling curve for pure substances

(b) **Mixture of two solids:** When the mixture of two liquids is cooled slowly and the cooling curves is constructed by plotting temperature against time, a cooling curve shown in Fig. 2.5 is obtained. Point b marks the beginning of freezing or appearance of solid. However the temperature does not remain constant. The temperature decreases continuously, but at a different rate (compare with cooling curve of a pure substance). Actually at point b one of the components begins to solidify or crystallize. During this solidification process, heat is liberated and therefore, the temperature falls more slowly. From point b to c as the solidification proceeds, the composition of the mixture changes continuously. At a certain temperature T, known as *eutectic temperature*, the remaining liquid solidifies as a whole, i.e. both the components crystallise out separately. The temperature remains constant from point *c* to *d*. This is also known as *temperature halt*.

(ii) **Eutectic Point:** The term eutectic implies 'easy melting'. A binary system consisting of two substances, which are miscible in all proportions in the liquid phase and do not react chemically, is known as **eutectic system**. The two substances involved have the property of lowering each other's freezing point. Further, solid solution of two or more substances having the **lowest freezing point** of all the possible mixtures of the components is known as **Eutectic mixture**. The minimum freezing point attainable corresponding to the eutectic mixture, is termed as the **eutectic point** (lowest melting point). The eutectic mixture

2.7. The percentages would then be given by the lever rule :

Two component system

$$\% \text{ crystals of A} = b/(a + b) \times 100$$

$$\% \text{ liquid} = a/(a + b) \times 100$$

SILVER-LEAD SYSTEM

It is an example of simple eutectic system in which the pure components alone separate as solid phases on cooling a molten mixture. Silver and lead in the solid state do not form intercomponent compounds because these are immiscible in solid state and do not react chemically. When molten silver and molten lead are mixed together in all proportions, a single homogeneous solution is formed. The system consists of four phases :

(i) solid silver (ii) solid lead
(iii) solution of molten silver and lead (iv) vapour

Since, the system is studied at constant pressure, the vapour phase can be ignored and the condensed phase rule is applicable

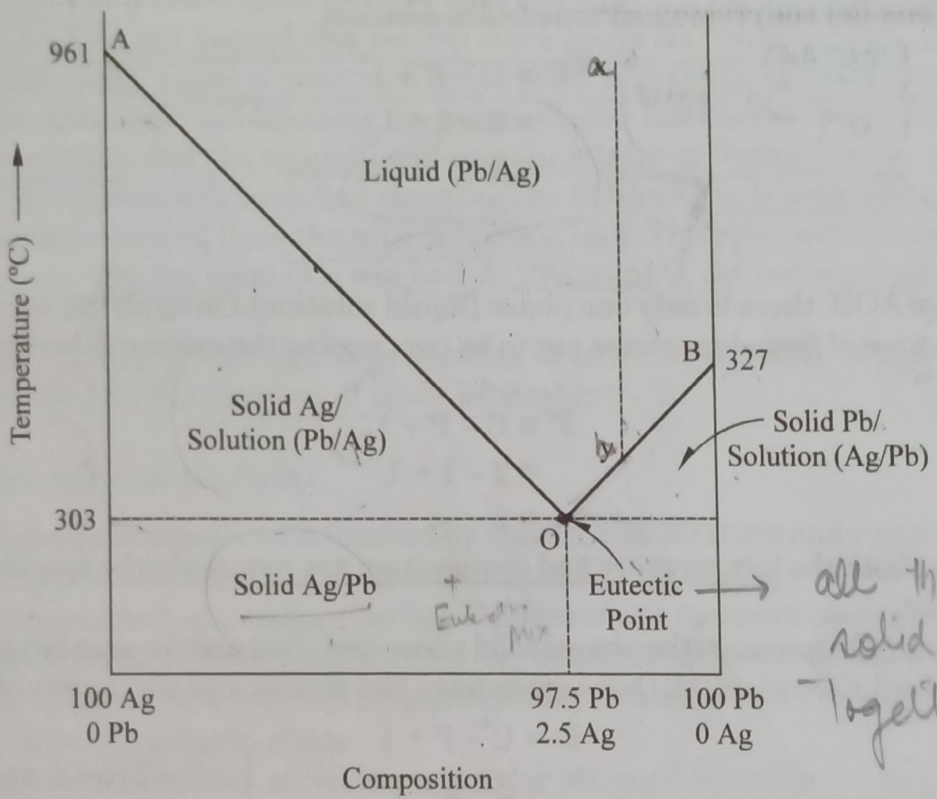
$$F' = C - P + 1$$

Imp

The maximum number of phase (when F = 0) that exist for this system is:

$$P = C - F' + 1$$
$$= 2 - 0 + 1$$
$$\boxed{= 3}$$

The phase diagram of lead silver system is shown in Fig. 2.8.



all three phases solid A, solid B and Liq can exist Together

Fig. 2.8 The phase diagram of silver-lead system

The phase diagram consists of the following :

(a) *Points*
Point A represents – melting point of pure silver (961°C).
Point B represents – melting point of pure lead (327°C).
Point O – eutectic point.

(b) *Curves*
Curve AO – melting point curve of silver in presence of lead
Curve BO – melting point of lead in presence of silver.

(c) *Areas*
Area above AOB.
Area below AO.
Area below OB.

Points

In the phase diagram point A represents the melting point of pure silver and b that of pure lead. It is clear from the diagram that adddition of Pb to pure Ag lowers the melting point of Ag. Similarly addition of Ag to pure Pb lowers the melting point of Pb. All the points on the curves AO and BO, represent the melting point of various mixtures of Ag and Pb.

Curves

Curve AO is the melting point curve of silver, it shows the effect of addition of Pb on the melting point of pure Ag. Along this curve solid Ag and solution are in equilibrium.

Curve BO is the melting point curve of lead. It shows the effect of addition of Ag on the melting point of Pb. Along this curve solid Pb and solution are in equilibrium.

Hence, AO and BO both represent univariant systems,

C=2 (Pb, Ag)
P=2 (Liq and solid)

$$F' = C - P + 1$$
$$= 2 - 2 + 1$$
$$= 1$$

Areas

In the area above AOB, there is only one phase (liquid solution).On applying reduced phase rule equation, the degree of freeedom comes out to be two, means the system is bivariant in the area above AOB.

$$F' = C - P + 1$$
$$= 2 - 1 + 1$$
$$= 2$$

It means that both the temperature and composition are required to be specified to define the system completely.

The area below AO represents the phases solid silver + solution and the area below BO represents the phases solid lead + solution. All these areas have two phases and one degree of freedom.

$$F' = C - P + 1$$
$$= 2 - 2 + 1$$
$$= 1$$

$\left[\because C = 2 \text{ (lead and silver)}\right.$
$\left. P = 2 \text{ (liquid and solid phase)}\right]$

The system is univariant in areas below AO and OB.

Eutectic Point O

The two curves AO and BO intersect at point O at a temperature 303°C. The point O is known as eutectic point or eutectic temperature.

At this point three phases solid silver, solid lead and the melt are in equilibrium. On applying phase rule equation to this point we get,

C = Pb, Ag
P = Pb(s) Ag(s) and Melt

$$F' = C - P + 2$$
$$= 2 - 3 + 1$$
$$= 0$$

Thus the system solid silver/solid lead/solution of both Ag and Pb is monvariant at point O. At this point both temperature (303°C) and composition (Ag = 2.6% and Pb = 97.4) are fixed. If the temperature is increased above the eutectic temperature, the solid phase (silver and lead) will disappear and if the temperature is decreased below the eutectic point, solution phase will disappear and only solid phase (silver and lead) will remain in existence.

The mixture (Ag = 2.6% and Pb = 97.4%) at this point is called as eutectic mixture. The eutectic mixture has a definite melting point, but it is not regarded as a compound because

(i) The components are not in stoichiometric proportion.

(ii) The existence of separate crystals of the components can be observed on examining it under microscope.

alloy

Application of phase diagram of two component system or Pattinson's process for the desilverisation of argentiferrous lead

The phase diagram of silver-lead system has a special significance in relation to disilverisation of Lead of argentiferrous lead. The argentiferrous lead contains very small amount of silver. This argentiferrous lead is first heated to a temperature above its melting point. Thus the system consists of only the liquid phase represented by the point '*a*' in the figure. It is then allowed to cool, the temperature of the melt will fall along the perpendicular line *ab*. As the point '*b*' is reached lead will begin to crystallise and the solution will contain relatively increasing amount of silver. On cooling further, the system will be shifted along the line BO, lead starts separating out continuously and it is constantly removed from the argentiferrous lead. Thus the melt continues to be richer and richer in silver until the point O is reached. At this point O, the percentage of silver increases to 2.6% by mass.

The process of raising the relative proportion of silver in the alloy is called as Pattinson's Process. This is used for the recovery of silver profitably.

Characterstics of Eutectic Point

1. It is an invariant piont and is marked by specific temperature and composition.
2. It is the lowest melting point of any mixture of Ag and Pb.
3. Below the eutectic temperature, no liquid phase of any composition can exist in the system.
4. If the liquid melt of the two components (silver and lead) is cooled below the eutectic point, both the components solidify simultaneously without any change in composition or temperature of the liquid phase.
5. The eutectic mixtures of alloys have greater strength than their components because of their crystal charaterstics.
6. An eutectic system can maintain its temperature contant over long peroids.

Use of Eutectic System

1. The eutectic systems are used in the preparation of low melting alloys such as various types of solders useful in metal joints and electric fuses.
2. An alloy 50% Bi + 25% Pb + 12.5% Sn + 12.5% Cd is known as wood metal alloy. It has a low melting piont of 65°C. This is used for plugs in water sprayers in buildings. In case of fire accident, the plugs melt away and water is released automatically to extinguish fire.
3. Freezing mixutures: A system at the eutectic point can maintain its temperature over long periods and such systems are used as constant temperature baths. The lowest possible temperature attainable with a certain freezing mixture depends upon the eutectic temperature of the system. Some common freezing mixtures are given in the Table 2.1.